

Personal portfolio website hosted on an Azure Linux Cloud server

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GitHub Repository: <https://github.com/choningy20-pixel/ICT-171-Cloud-Project>

Server IP Address: 20.58.179.128

Purpose:

The purpose of this project is to create and host a simple personal portfolio website using an Azure Linux Virtual Machine. The website will be accessible through a public IP address and later connected to a domain name. The project will demonstrate cloud server setup, Apache server installation, basic Linux command usage, DNS configuration, GitHub documentation and a simple script.

Cloud Services Used:

- ❖ Microsoft Azure
- ❖ Azure Virtual Machine
- ❖ Azure Bastion
- ❖ Ubuntu Linux
- ❖ Apache2 Web Server
- ❖ GitHub

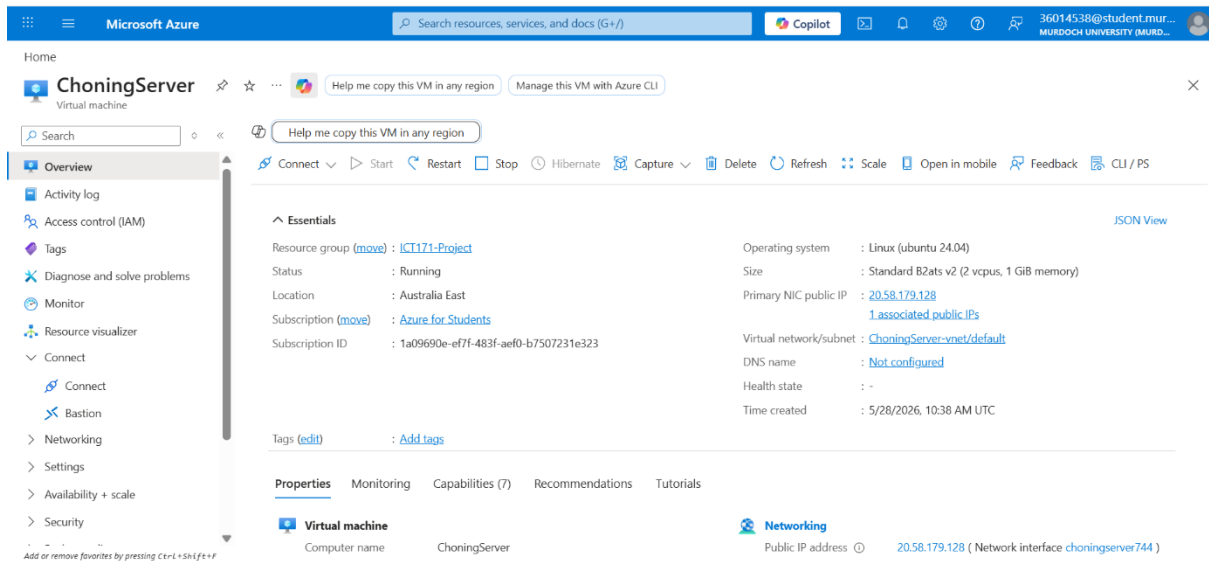


Figure 1: Azure Virtual Machine Overview showing the running Ubuntu Linux VM hosted on Microsoft Azure.



Figure 2: Personal portfolio website successfully hosted online using Apache2 web server.

```
Outcome:
The website was successfully hosted online and accessible through the public IP address of the Azure virtual machine.
azureuser@ChoningServer:~$ ystemctl status apache2
Command 'ystemctl' not found, did you mean:
  command 'systemctl' from deb systemd (255.4-1ubuntu8.15)
  command 'systemctl' from deb systemctl (1.4.4181-1.1)
Try: sudo apt install <deb name>
azureuser@ChoningServer:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-28 10:52:57 UTC; 1h 18min ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 2691 (apache2)
      Tasks: 55 (limit: 977)
     Memory: 6.5M (peak: 6.9M)
        CPU: 249ms
    CGroup: /system.slice/apache2.service
            └─2691 /usr/sbin/apache2 -k start
              └─2694 /usr/sbin/apache2 -k start
                └─2695 /usr/sbin/apache2 -k start

May 28 10:52:57 ChoningServer systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 28 10:52:57 ChoningServer systemd[1]: Started apache2.service - The Apache HTTP Server.
azureuser@ChoningServer:~$
```

Figure 3: Apache2 web server status showing active and running service.

```
└─2694 /usr/sbin/apache2 -k start
└─2695 /usr/sbin/apache2 -k start

May 28 10:52:57 ChoningServer systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 28 10:52:57 ChoningServer systemd[1]: Started apache2.service - The Apache HTTP Server.
azureuser@ChoningServer:~$ ./server-status.sh
===== Server Status Report =====
Hostname:
ChoningServer

Current date and time:
Thu May 28 12:12:32 UTC 2026

System uptime:
12:12:32 up 1:33, 1 user, load average: 0.00, 0.00, 0.00

Disk usage:
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        29G   1.9G   27G   7% /
tmpfs            422M    0   422M   0% /dev/shm
tmpfs            169M  988K   168M   1% /run
tmpfs             5.0M    0    5.0M   0% /run/lock
efivarfs         128K   38K   86K   31% /sys/firmware/efi/efivars
/dev/sda16       881M   64M   756M   8% /boot
/dev/sda15       105M   6.2M   99M    6% /boot/efi
tmpfs             85M   12K   85M    1% /run/user/1000
azureuser@ChoningServer:~$
```

Figure 4: Linux shell script output displaying server status information.

Conclusion

This project successfully demonstrated the setup and deployment of a cloud-based Linux web server using Microsoft Azure. An Ubuntu virtual machine was created and configured with Apache2 web server software. A custom HTML website was hosted online and accessed through the VM's public IP address. Linux terminal commands and shell scripting were also used to monitor server information and system status. Overall, the project provided practical experience in cloud computing, Linux server administration, web hosting, and basic scripting.

References

Microsoft Azure Documentation

<https://learn.microsoft.com/>

Ubuntu Server Documentation

<https://ubuntu.com/server/docs>

Apache HTTP Server Documentation

<https://httpd.apache.org/docs/>